MUHAMMAD ANAS

AI Automation & AI Agents | Generative AI | Computer Vision | Full-Stack Development

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SUMMARY

Computer Science student with hands-on experience building end-to-end AI applications, multi-agent systems, and full-stack platforms. Skilled in training machine learning models, computer vision forensics, and integrating LLMs/RAG architectures into production-ready web apps. Passionate about turning complex business logic into intelligent, autonomous systems that deliver measurable impact.

TECHNICAL SKILLS

Programming Languages:	Python, JavaScript, C++, C, Java , Dart , C#
Generative AI:	LLMs, RAG, LangChain, Multi-Agent Systems, Agentic AI, Prompt Engineering, Gemini API, FAISS, Semantic Search
ML & Computer Vision:	PyTorch, XGBoost, Scikit-Learn, OpenCV, Feature Engineering, Classification Models, Model Evaluation, Computer Vision Forensics
Backend & APIs:	Node JS , Express JS , FastAPI, Django , REST APIs, Async Programming, API Integration, Webhooks
Frontend:	HTML, CSS , React.js, Next.js, Streamlit, Responsive Design, UI Integration
Databases:	PostgreSQL, MongoDB, MySQL, Vector Databases, Database Design, Query Optimization
AI-Assisted Development:	Claude Code, OpenAI Codex, Antigravity, Cursor
Tools	Git, GitHub, Docker, Postman, VS Code, Figma, Vercel, Netlify
Data Processing:	Data Cleaning, Data Validation, Feature Extraction, Data Preprocessing, PDF Processing, Document Intelligence

PROJECTS

Agentic Clinical Intelligence Partner

Python, XGBoost, PyTorch, FAISS, Gemini 2.0, Streamlit

- Multi-agent emergency department triage system using a sequential pipeline of 10 distinct agents to process patient data and generate treatment recommendations.
- Trained a 5-class XGBoost classifier on 87,000+ records achieving 89.97% accuracy; integrated KNN and K-Means for patient similarity and profiling.
- Developed an Imaging Agent (DenseNet-121 + CLAHE + GradCAM) for automated X-ray analysis across 18 pathologies with visual heatmaps.
- Implemented a RAG pipeline (FAISS + Gemini) and real-time monitoring dashboard to generate evidence-based treatment plans and predict patient deterioration.

Payment Screenshot Fraud Detection

Python, OpenCV, Scikit-Learn, XGBoost, Streamlit

- Hybrid forensic-grade pipeline to detect forged mobile payment receipts, achieving 93%+ accuracy through pixel-level analysis and machine learning.
- Engineered 128 custom forensic features using ELA, LBP (texture), and DCT variance for deep image authenticity analysis.
- Developed a two-tier decision engine with local XGBoost for fast inference and Gemini 2.5 Pro fallback for uncertain cases.
- Created an interactive Streamlit app with ROI extraction, ELA heatmaps, confidence scoring, and downloadable forensic reports.

RAG-based Document AI Assistant

LangChain, Google Gemini, Python

- Developed a Question Answering system for intelligent document extraction and context-aware reading over dense PDFs.
- Engineered retrieval-augmented generation pipelines enabling semantic search and accurate synthesis for end-user text queries.

EXPERIENCE

Web Developer Intern · *CIMT* — Jan 2025 – Feb 2025 | Mansehra, Pakistan

- Worked with UI/UX designs and converted them into responsive web pages using React.js and CSS.
- Followed design layouts to maintain a clean, user-friendly, and visually consistent interface.
- Designed and organized MongoDB database schemas for storing user and application data.
- Ensured smooth data flow between frontend and backend so the application functions correctly.

EDUCATION

National University of Computer and Emerging Sciences (FAST-NUCES) — 2023 – Ongoing
Bachelor of Science in Computer Science — *Peshawar, Pakistan*

Relevant Courses: OOP, Data Structures & Algorithms, Database Systems, Operating Systems, Software Design, AI

Peace Group of School and Colleges — 2020 – 2022
FSC Pre-Engineering – 89% — *Mansehra, Pakistan*

Peace Group of School and Colleges — 2018 – 2020
Matriculation (Science) – 85% — *Mansehra, Pakistan*